Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30-09-26
Team ID	23KD1A4230
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements

The following functional requirements define the core features, user interactions, and specific system behaviours of the proposed digital metro ticketing solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Access & Portal Navigation	Access Service Catalog via mobile browser Commuter login/guest booking authentication
FR-2	Journey Configuration	Select Source and Destination stations from active lists Input passenger count and ticket type
FR-3	Dynamic Fare Calculation	Auto-calculate base fare based on station distance Apply dynamic pricing rules (e.g., peak hours)
FR-4	Ticket Generation & Processing	Create backend ServiceNow request (REQ/RITM) Generate unique scannable digital QR code ticket
FR-5	Notifications & Alerts	Send instant booking confirmation via email Send digital ticket retrieval link via SMS
FR-6	Ticket Validation	Authenticate QR code scan Update backend ticket status to "Used/Completed"
FR-7	Administration & Reporting	Generate daily revenue and booking volume dashboards Export station traffic and bottleneck statistics

Non-functional Requirements:

The following non-functional requirements specify the critical quality attributes, performance standards, and security constraints required for the system to operate effectively.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Service Portal interface must be highly intuitive, visually clean, and strictly mobile-responsive to accommodate commuters booking on their phones.
NFR-2	Security	All commuter travel history and payment data must be securely restricted using role-based Access Control Lists (ACLs) to prevent unauthorized data exposure.
NFR-3	Reliability	The system must guarantee that a generated QR code maps perfectly to the backend database every time, preventing commuters from being stranded at validation gates.
NFR-4	Performance	The catalog form must load in under 2 seconds, and the client scripts calculating the dynamic fare must execute instantly without freezing the user interface.
NFR-5	Availability	The ticketing portal must maintain a 99.9% uptime, ensuring uninterrupted service access during all operational metro hours, including weekends and holidays.
NFR-6	Scalability	The application backend and database structure must be capable of processing severe, sudden spikes in concurrent user traffic during morning and evening rush hours without lagging.